

Introduction to Transparency-oriented Visualizations

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Lecture 3 goals



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How does transparency relate to data visualization?

Theoretical background

How to design (transparent) visualizations?

R tutorial using ggplot2, tidybayes, and ggdist

How to make choices that increase the transparency and usability in visualizations?

Practical guidelines

**What are transparency-
oriented visualizations?**

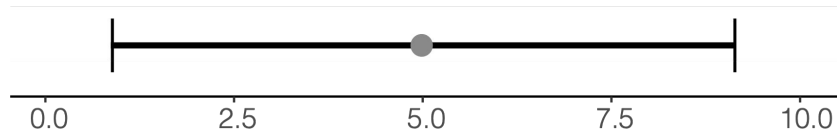
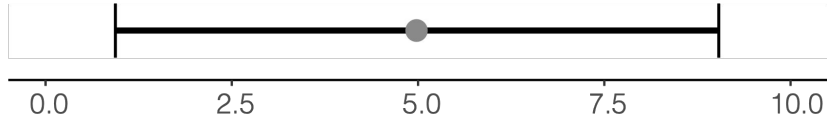
Expressiveness is a proxy to transparency

Expressiveness principle: the visual representation should represent *all* and *only* the relationships that exist in the data
(Mackinlay, 1986, Munzner, 2014)

[Mackinlay, J. \(1986\)](#). Automating the design of graphical presentations of relational information.

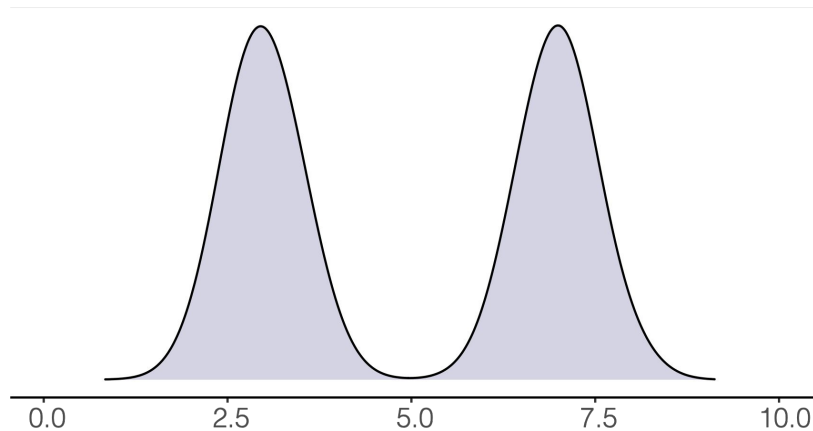
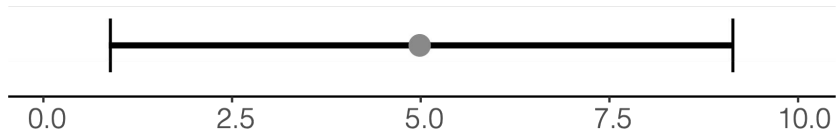
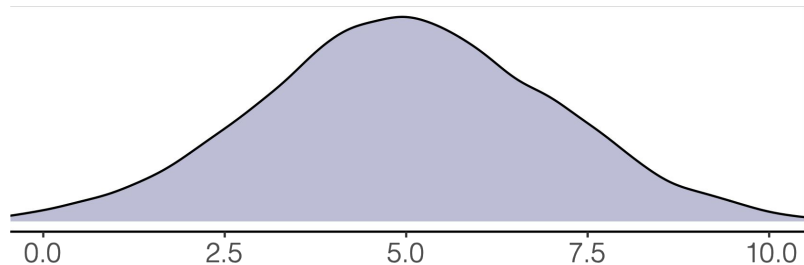
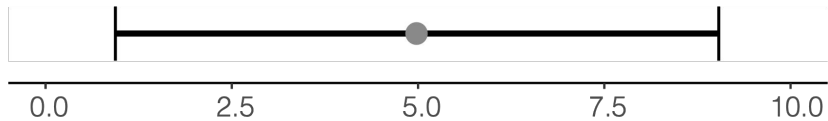
[Munzner, T. \(2014\)](#). Visualization analysis and design. CRC press.

What can you say about these two 95% confidence intervals?



Intervals (95% CI)

Summaries can **obscure** important relationships in distributional data



Intervals (95% CI)

Distributions

Expressiveness is a proxy to transparency

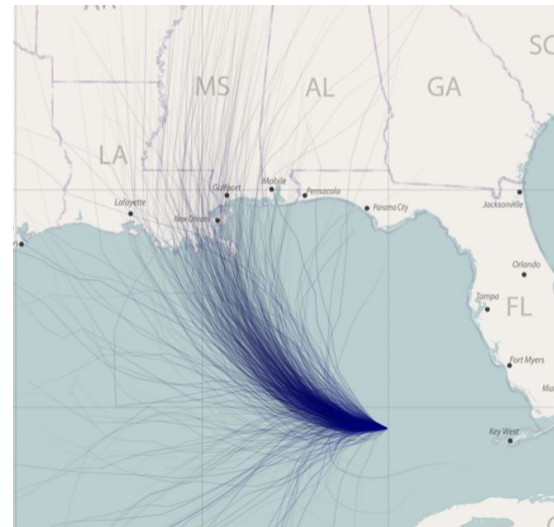
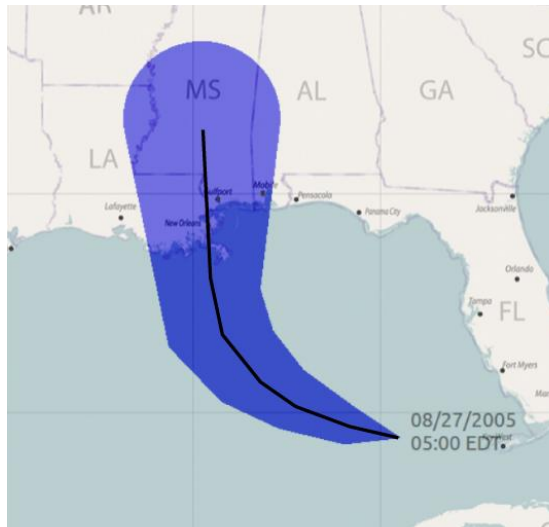
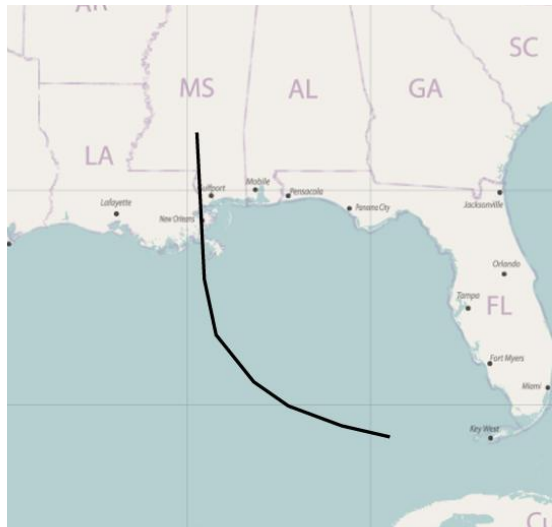
Expressiveness principle: the visual representation should represent *all* and only the relationships that exist in the data (Mackinlay, 1986, Munzner, 2014)

Visualizations of data can vary on a spectrum of expressiveness.

[Mackinlay, J. \(1986\)](#). Automating the design of graphical presentations of relational information.

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Visualizations of data can vary on a spectrum of expressiveness



expressiveness —————→

Expressiveness is a proxy to transparency

Expressiveness principle: the visual representation should represent *all* and only the relationships that exist in the data (Mackinlay, 1986, Munzner, 2014)

But, expressiveness can trade off with usability.

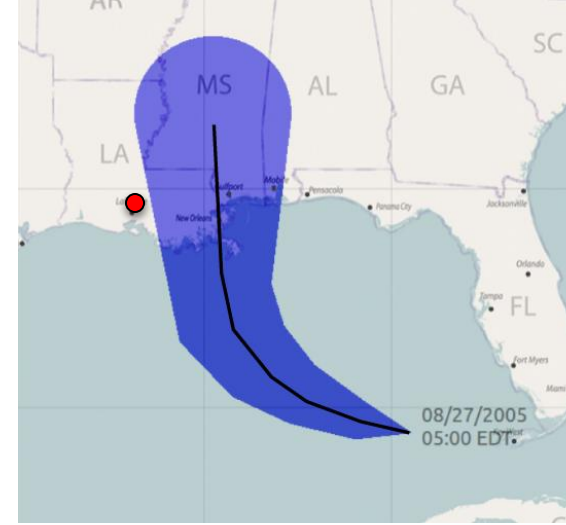
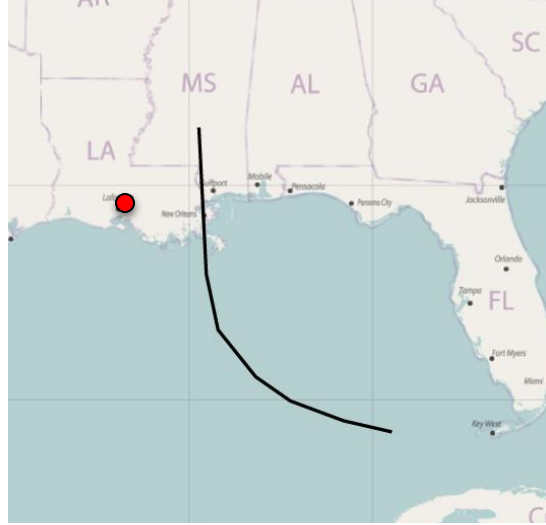
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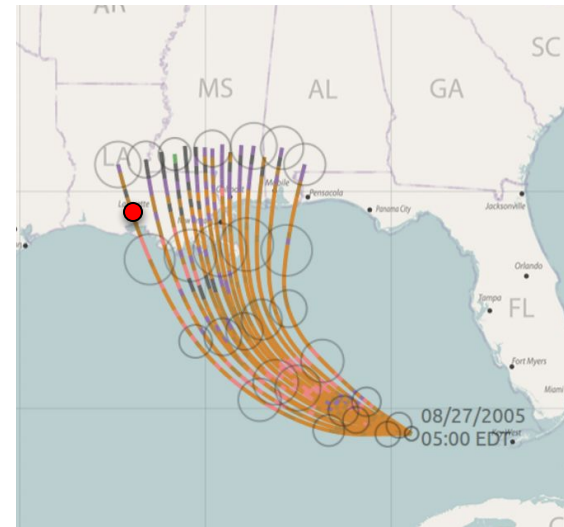
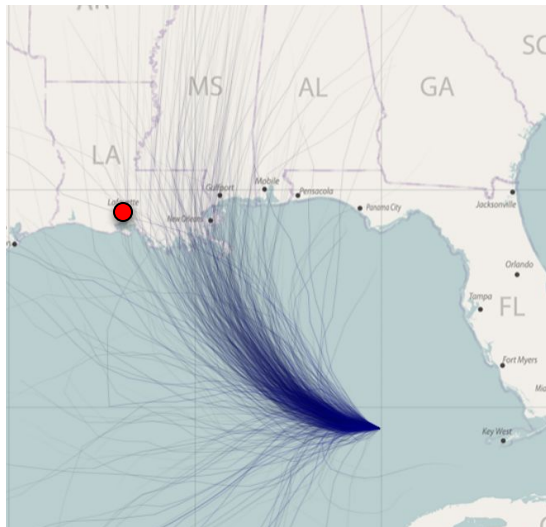
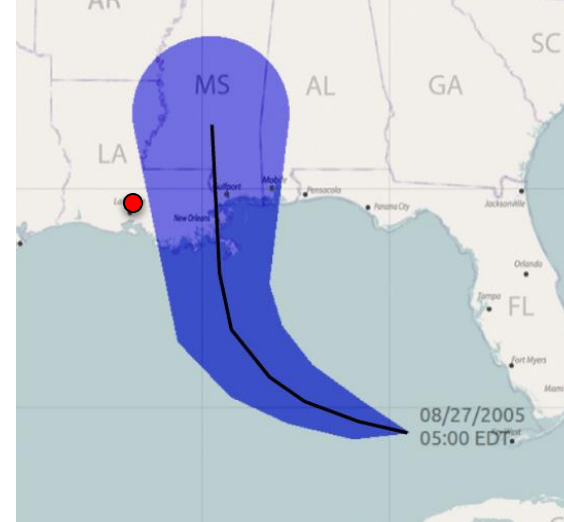
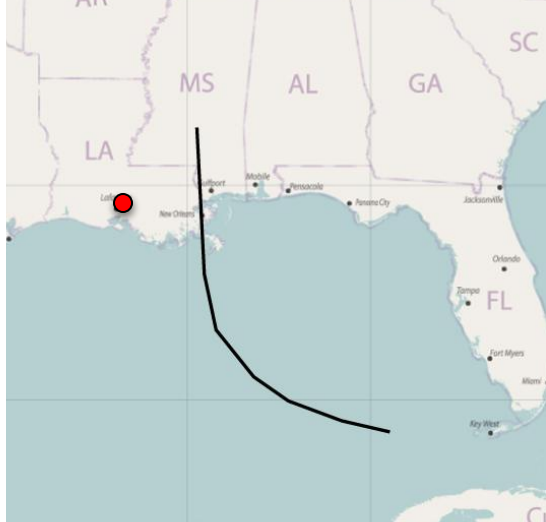
Goals of Transparency-oriented Visualization

1. Faithfully represent the data: **Expressiveness**
2. Support users in making accurate inferences from the data: **Usability**

Would you stay or evacuate?



Would you stay or evacuate?



What are transparency-oriented visualizations?

intelligent decisions about tradeoffs between expressiveness and usability

Importance of uncertainty

Without uncertainty, viewers may come to **incorrect** conclusions about the data.

Why visualizing uncertainty matters

Scientific Credibility and Accuracy

Nosengo, N. (2014) Italian seismologists fight to overturn convictions.

nature

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Published: 20 October 2014

Italian seismologists fight to overturn convictions

[Nicola Nosengo](#)

[Nature](#) (2014) | [Cite this article](#)

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Six scientists and one official contend that they never reassured the public before deadly 2009 L'Aquila quake.



Why visualizing uncertainty matters

Trust

“Our aim – everybody’s aim – is surely to trust the trustworthy, but not the untrustworthy.”

Onora O'Neill (2012)

Representing uncertainty supports accurate calibrations of trust.

Why visualizing uncertainty matters

People expect uncertainty

**Researchers find that people
assume there is uncertainty**
(Joslyn & Savelli, 2010)

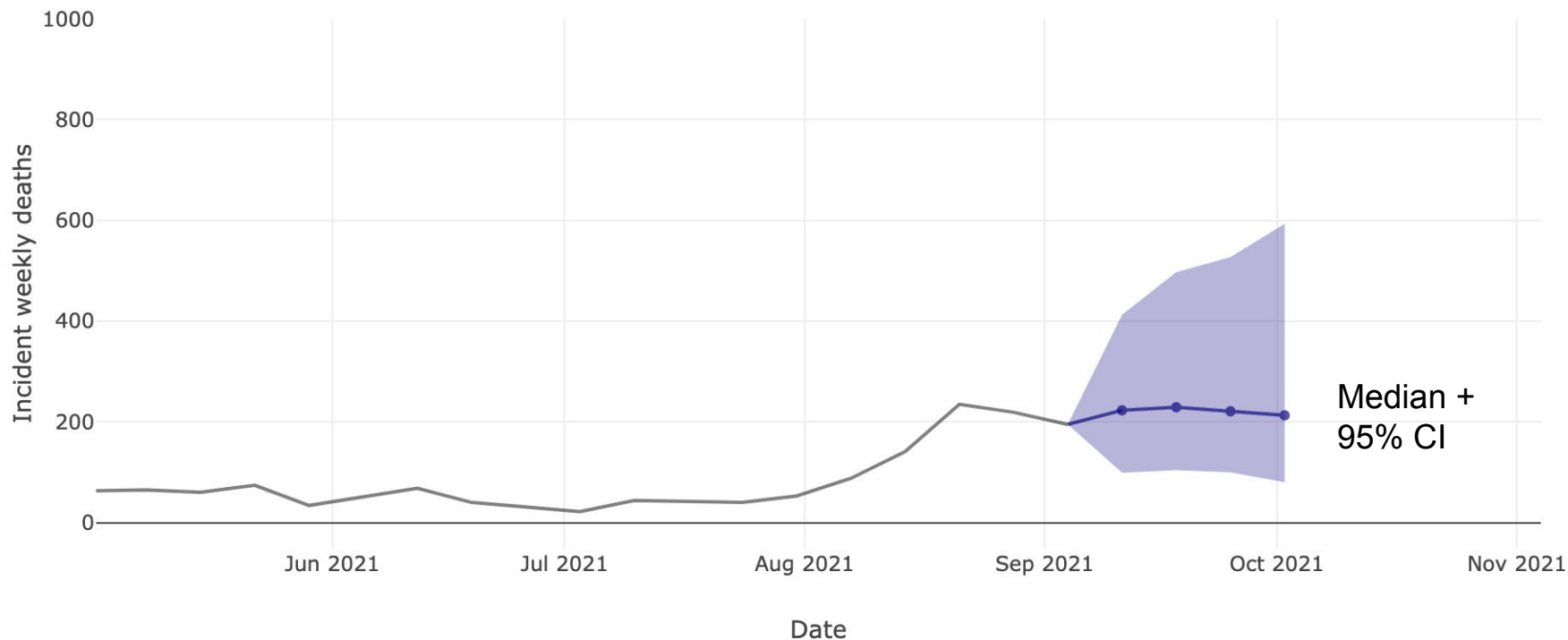
Representing the uncertainty
grounds their assumptions in
the data

Joslyn, S., & Savelli, S. (2010). Communicating forecast uncertainty: Public perception of weather forecast uncertainty. *Meteorological Applications*, 17(2), 180-195.

Importance of uncertainty

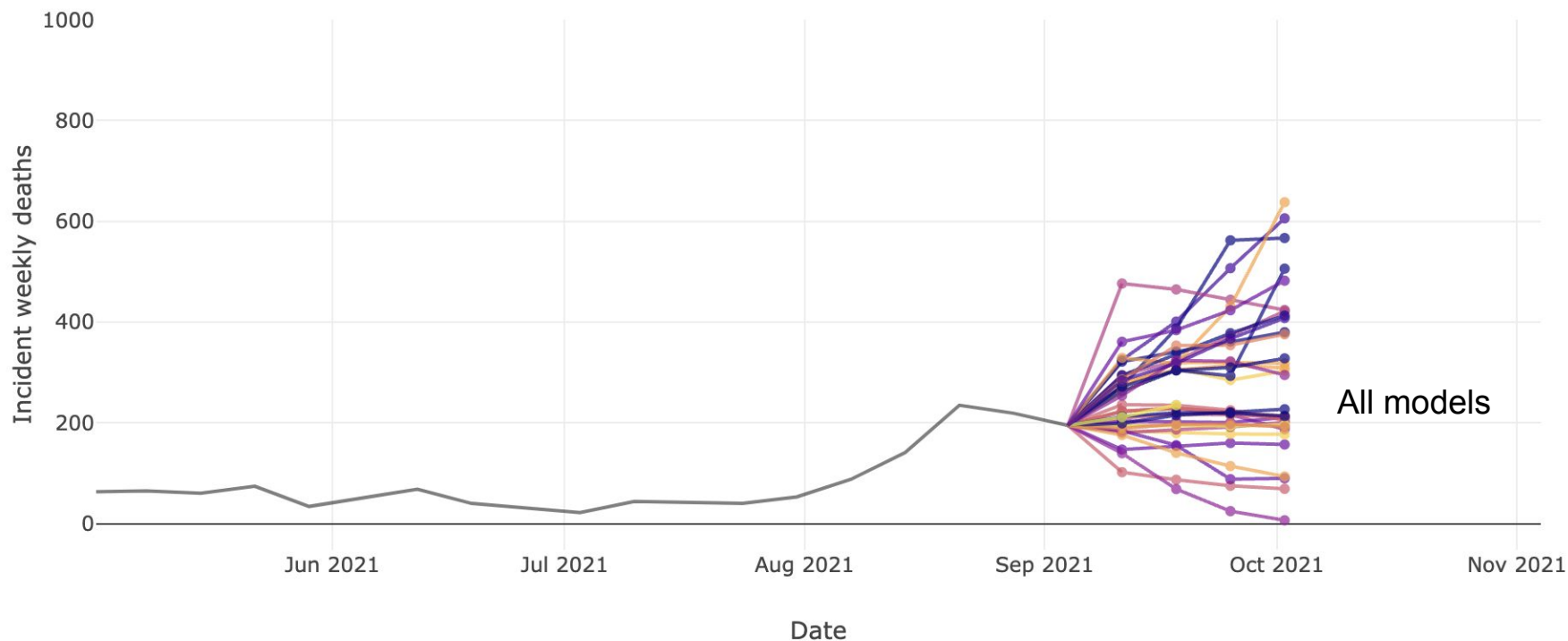
intelligent decisions about **tradeoffs** between
expressiveness and usability

Forecasts of Incident weekly deaths in Alabama as of 2021-09-04



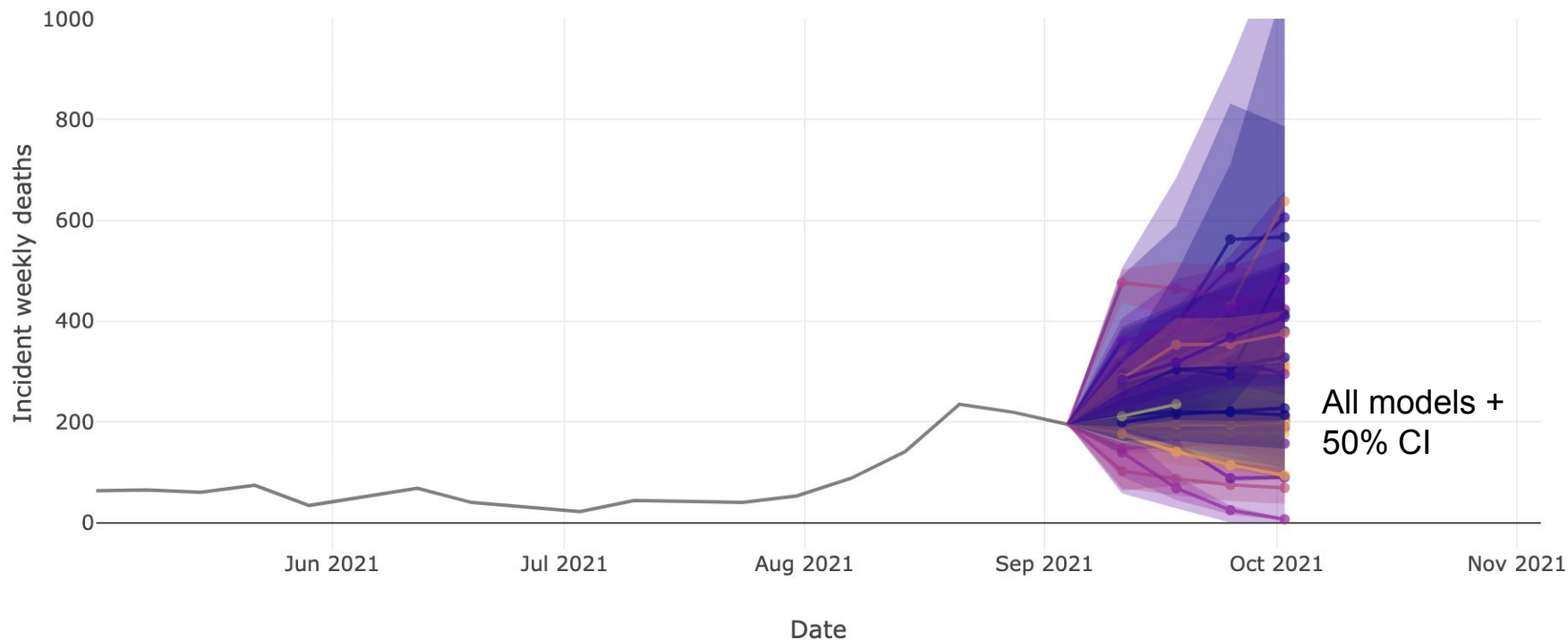
expressiveness
usability

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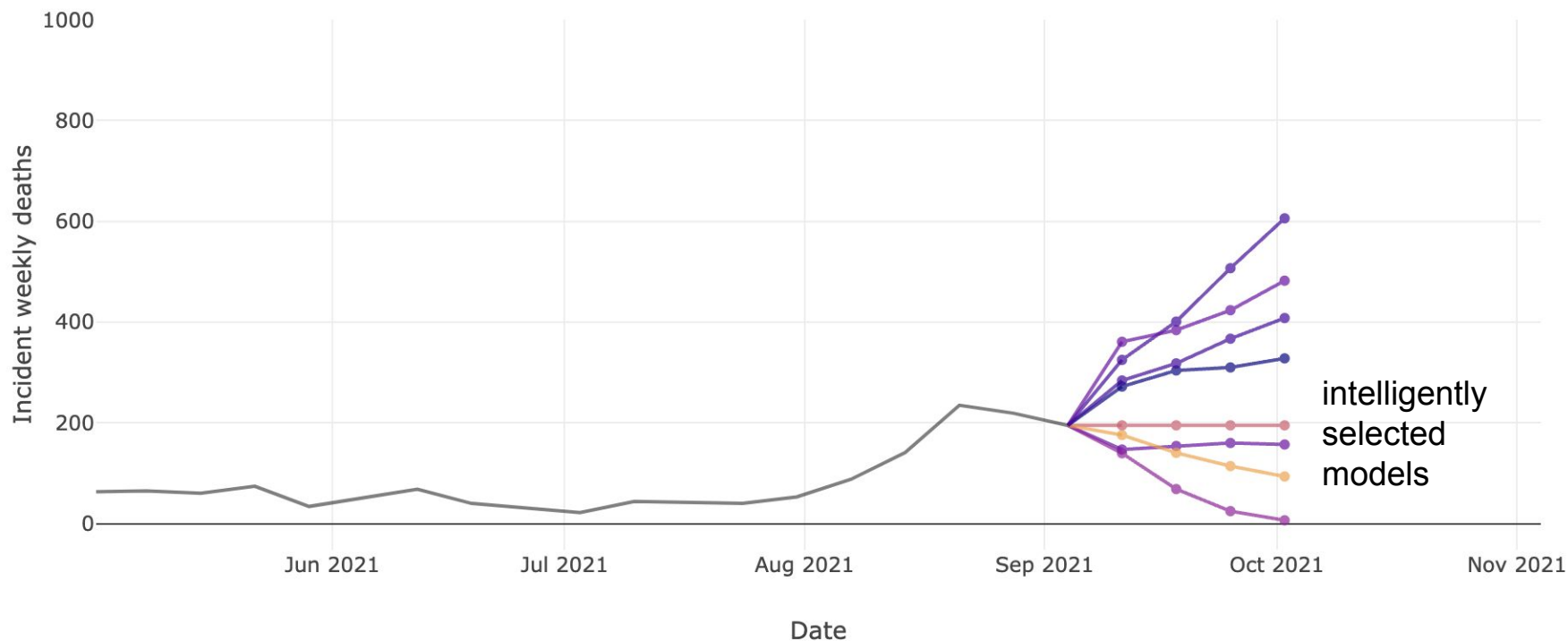
expressiveness
usability

Forecasts of Incident weekly deaths in Alabama as of 2021-09-04



expressiveness
usability

Forecasts of Incident weekly deaths in Alabama as of 2021-09-04



expressiveness
usability

Intelligent decisions about tradeoffs between expressiveness and usability

These decisions will be different depending on the context, data set, and audience

Lecture 3 Goals

How does transparency relate to data visualization?

Theoretical background

How to design (transparent) visualizations?

R tutorial using ggplot2, tidybayes, and ggdist

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Practical guidelines

Begin tutorial

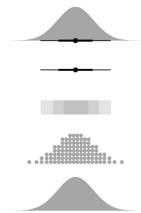
[5-min Break] Pull up the Rstudio project

Recap & design tips

- Expressiveness
- Usability

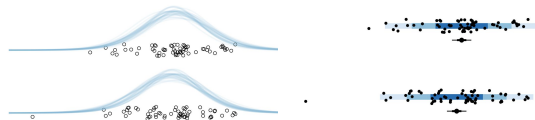
You also learned about:

- Types of uncertainty
- Uncertainty representations



Tips and resources

- Add raw data to the modeling results
- Label things directly
- Try alternatives and evaluate
- `ggdist`



Please answer a one-question survey about this session:
<http://tiny.cc/t-chi23-feedback-3>

Homework

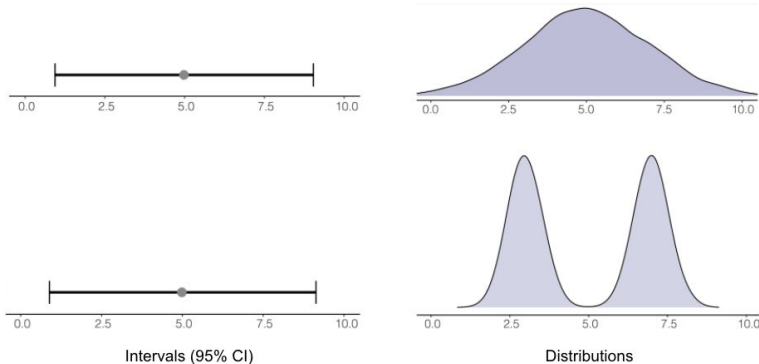
Choose your past paper* improve its transparency practices by applying one or more methods learned in lectures 1–3

Prepare a max. 10-minute presentation of your work in any format

In Lecture 4 (April 29), we will break into several groups to see your presentations. Peers and instructors will give feedback

*If you don't have any previously published work, let us know. We will give you one of CHI papers from the '90s

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Scientific Credibility and Accuracy

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Italian seismologists fight to overturn convictions

Nicola Nosenno

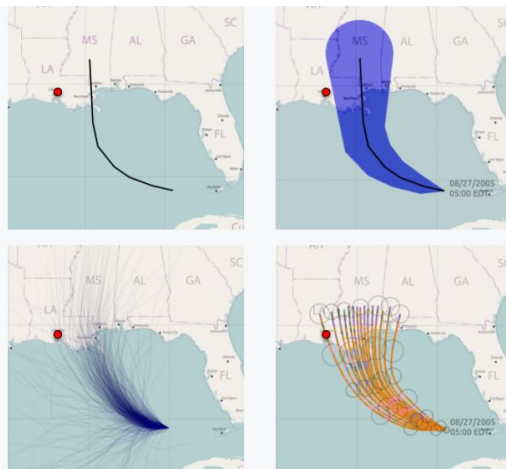
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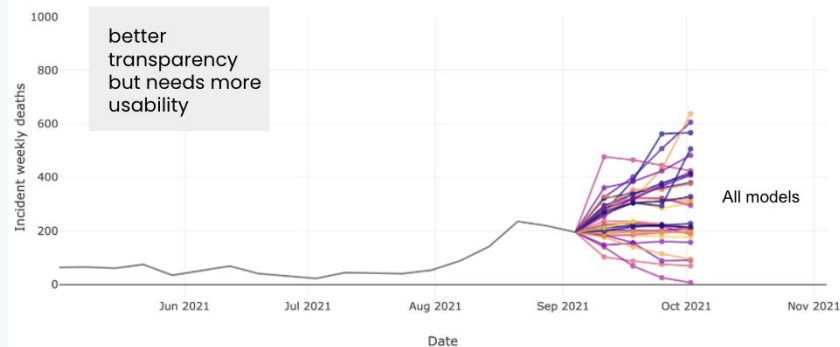
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Would you stay or evacuate?



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